

Probability and statistics

Background material

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What this deck is for

The course assumes a working knowledge of probability. This deck collects the notions it actually uses, and nothing more.

Notion	Where the course needs it
Probability space, measurability	$\mathcal{Q}(x)$ is well defined (deck 03)
Support Ξ	K_2 , K_2^P , scenario trees (decks 03, 06)
Quantile F^{-1}	inversion sampling (deck 02), chance constraints (deck 05)
Expectation, second-order moments	nearly every theorem of decks 03–04
Jensen's inequality	$EV \leq WS$ (deck 03)
“almost surely”	K_2 versus K_2^P (deck 03)
Conditional expectation	multistage, non-anticipativity (decks 06–08)
Modes of convergence, LLN	sample average approximation (deck 09)
Central limit theorem, CLs	stopping rules (decks 07, 10, 12)

Throughout, ξ denotes a random vector and ξ one of its realizations — the convention used in the slides.

Probability space

A **probability space** is a triplet $(\Omega, \mathcal{F}, P)$:

1. Ω is the **sample space**, an arbitrary non-empty set;
2. $\mathcal{F} \subseteq 2^\Omega$ is a **σ -algebra** (or σ -field): a collection of subsets of Ω , called **events**, such that
 - $\Omega \in \mathcal{F}$,
 - $A \in \mathcal{F} \Rightarrow \Omega \setminus A \in \mathcal{F}$,
 - $A_i \in \mathcal{F}, i = 1, 2, \dots \Rightarrow \bigcup_i A_i \in \mathcal{F}$;
3. $P : \mathcal{F} \rightarrow [0, 1]$ is a **probability measure**: $P[\Omega] = 1$, and $P[\bigcup_i A_i] = \sum_i P[A_i]$ for any countable collection of pairwise disjoint $A_i \in \mathcal{F}$.

An event A *occurs* if the outcome ω of the experiment belongs to A .

Why a σ -algebra?

Why not simply take $\mathcal{F} = 2^\Omega$, and give every subset a probability?

Because on an uncountable Ω — say $[0, 1]$ — no probability measure can assign a consistent value to *every* subset. One has to restrict attention to a family that is rich enough to be useful and small enough to be consistent. That family is a σ -algebra, and its members are exactly the sets we are allowed to call events.

Two consequences that matter for the course:

- a function of ω is only usable if it is **measurable** — otherwise “ $P[\xi \leq x]$ ” is meaningless;
- in deck 03, the measurability of $Q(x, \xi)$ is what makes $\mathcal{Q}(x) = \mathbb{E}_\xi[Q(x, \xi)]$ well defined at all. It is assumed, not proved, there.

For a finite or countable Ω , $\mathcal{F} = 2^\Omega$ causes no trouble, and this discussion can be ignored: that is the setting of most scenario-based models.

Random variables and random vectors

A **random variable** ξ on $(\Omega, \mathcal{F}, P)$ is a function $\xi(\omega)$, $\omega \in \Omega$, such that

$$\{\omega \mid \xi(\omega) \leq x\} \in \mathcal{F} \quad \text{for every } x \in \mathbb{R},$$

so that a probability can be attached to $\{\xi \leq x\}$. A **random vector** $\xi : \Omega \rightarrow \mathbb{R}^r$ is defined componentwise.

Its **cumulative distribution function** (cdf) is

$$F_{\xi}(x) = P[\xi \leq x].$$

The support

The **support** $\Xi \subseteq \mathbb{R}^r$ of ξ is the **smallest closed set** carrying all the probability, $P[\xi \in \Xi] = 1$. Equivalently, and more usable in practice,

$$\Xi = \{x \in \mathbb{R}^r \mid P[\xi \in V] > 0 \text{ for every open } V \ni x\} :$$

a point belongs to the support when no neighbourhood of it can be discarded.

That a *smallest* such set exists is what the second form guarantees: the complement of Ξ is a union of open sets of probability zero, and in $\mathbb{R}^r$ countably many of them already cover it, so the complement itself has probability zero.

The **closedness** is not a detail either. It is exactly what makes the argument of deck 03 work: there, the set of ξ favourable to a given x is closed and has probability one, and must therefore contain the whole support — which is how K_2 and K_2^P come to coincide under a fixed recourse. Note also that Ξ need not be connected, nor equal to the range $\xi(\Omega)$: a value taken with probability zero is not in the support.

Discrete and continuous random variables

Discrete: ξ takes finitely or countably many values ξ_k , with **probability masses** $p_k = P[\xi = \xi_k]$, $\sum_k p_k = 1$. This is the *scenario* setting of the course.

Continuous: ξ admits a **density** f , and then $P[\xi = x] = 0$ for every x , while

$$P[a \leq \xi \leq b] = \int_a^b f(x) dx = \int_a^b dF(x) = F(b) - F(a).$$

Writing dF rather than $f(x)dx$ covers both cases at once, and is the notation the slides use.

A distribution need be neither of the two: a discrete part and a density can coexist. Nothing in the course requires that generality, but the dF notation handles it.

The distribution function

For a random variable, F_ξ takes values in $[0, 1]$, is non-decreasing, and tends to 1 at $+\infty$ and to 0 at $-\infty$. It is moreover **right-continuous**, with left limits:

$$\lim_{y \downarrow x} F_\xi(y) = P[\xi \leq x] = F_\xi(x), \quad F_\xi(x^-) \stackrel{\text{def}}{=} \lim_{y \uparrow x} F_\xi(y) = P[\xi < x],$$

since the events $\{\xi \leq y\}$ decrease to $\{\xi \leq x\}$ in the first case, and increase to $\{\xi < x\}$ in the second. The **jump** of F_ξ at x is therefore the mass at x ,

$$F_\xi(x) - F_\xi(x^-) = P[\xi = x],$$

and F_ξ is continuous **exactly when ξ has no atom**.

- **Discrete**: $F_\xi(x) = \sum_{k: \xi_k \leq x} p_k$ increases only by jumps, of height p_k at each ξ_k . With finitely many scenarios, it is a **step function**: right-continuous, but not continuous.
- **With a density**: there is no atom, so F_ξ is continuous. The converse fails — a continuous cdf need not have a density — but the course never meets that case.

Quantiles

The **quantile function** is the **generalized inverse**

$$F_{\xi}^{-1}(p) = \inf\{x \in \mathbb{R} \mid F_{\xi}(x) \geq p\}, \quad p \in (0, 1),$$

defined whether or not F_{ξ} is invertible — in particular when F_{ξ} has jumps.

Two uses in the course.

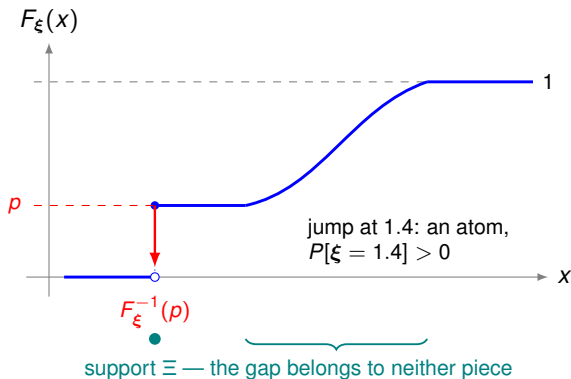
- **Inversion sampling** (deck 02). For $u \in (0, 1)$, $F_{\xi}^{-1}(u) \leq x \iff u \leq F_{\xi}(x)$, the infimum being attained by right-continuity, so $P[F_{\xi}^{-1}(u) \leq x] = F_{\xi}(x)$ for $u \sim U[0, 1]$ — **for every** F_{ξ} , discrete ones included.

Conversely, if ξ has no atom, $F_{\xi}(\xi) \sim U[0, 1]$: F_{ξ} is continuous, so $F_{\xi}(F_{\xi}^{-1}(u)) = u$ (right-continuity gives $\geq$, left-continuity $\leq$). F_{ξ} may still not be invertible, being constant across any gap of the support, but $\{F_{\xi}(\xi) \leq u\}$ and $\{\xi \leq F_{\xi}^{-1}(u)\}$ differ only on such a flat stretch, of probability zero:

$$P[F_{\xi}(\xi) \leq u] = P[\xi \leq F_{\xi}^{-1}(u)] = F_{\xi}(F_{\xi}^{-1}(u)) = u, \quad u \in (0, 1).$$

- A chance constraint $P[\cdot] \geq \alpha$ is a statement about a quantile (deck 05).

Cdf, quantile and support, graphically



A flat stretch of F is a region the variable never visits; a jump is an atom. The quantile is read horizontally then downwards, the inf picking the left end of a flat stretch. The support is what is left, closed up — here **disconnected**.

Expectation

$$\mathbb{E}[\xi] = \sum_k p_k \xi_k \quad (\text{discrete}), \quad \mathbb{E}[\xi] = \int_{-\infty}^{+\infty} x f(x) dx = \int_{-\infty}^{+\infty} x dF(x) \quad (\text{continuous}).$$

More useful in practice: for a measurable g ,

$$\mathbb{E}[g(\xi)] = \int_{\Xi} g(x) dF(x),$$

without having to know the distribution of $g(\xi)$. This is what allows the course to write $Q(x) = \mathbb{E}_{\xi}[Q(x, \xi)]$ directly.

Expectation is **linear**: $\mathbb{E}[a\xi + b\eta] = a\mathbb{E}[\xi] + b\mathbb{E}[\eta]$, whether or not ξ and η are independent. It is **monotone**: $\xi \leq \eta$ a.s. $\Rightarrow \mathbb{E}[\xi] \leq \mathbb{E}[\eta]$.

Both are used constantly and silently in the decks: linearity to split $\mathbb{E}[c^T x + q^T y]$, monotonicity to take expectations of an inequality, as in the proofs of $\text{EVPI} \geq 0$ and $\text{VSS} \geq 0$.

Moments

For an integer $p \geq 1$, the **moment of order p** of a random variable ξ is $\mathbb{E}[\xi^p]$, and its **central moment** of order p is $\mathbb{E}[(\xi - \mathbb{E}[\xi])^p]$: the mean is the first moment, the variance the second central moment. For any real $p > 0$, the **absolute moment** of order p is $\mathbb{E}[|\xi|^p]$.

ξ has **finite moments of order p** when $\mathbb{E}[|\xi|^p] < \infty$, which makes $\mathbb{E}[\xi^p]$ well defined and finite; for $p = 1$, ξ is said to be **integrable**. A random vector has finite moments of order p when $\mathbb{E}[\|\xi\|^p] < \infty$. For $p = 2$, as $\|\xi\|^2 = \sum_i \xi_i^2$, this asks every component to have a finite second moment: it is the hypothesis “ **ξ has finite second-order moments**”, which appears eight times in deck 03 alone.

Lower orders come for free. If $0 < r \leq p$, then $|x|^r \leq 1 + |x|^p$ for every x (bound $|x|^r$ by 1 if $|x| \leq 1$, by $|x|^p$ otherwise), hence

$$\mathbb{E}[|\xi|^r] \leq 1 + \mathbb{E}[|\xi|^p].$$

In particular, finite second-order moments give $\mathbb{E}[|\xi|] < \infty$: the mean exists and is finite. The converse fails: with $P[\xi = k] = c/k^3$, $k = 1, 2, \dots$, the mean $c \sum_k 1/k^2$ is finite, but $\mathbb{E}[\xi^2] = c \sum_k 1/k = +\infty$.

Variance and covariance

The **variance** is the second central moment,

$$\text{Var}[\xi] = \mathbb{E} [(\xi - \mathbb{E}[\xi])^2] = \mathbb{E}[\xi^2] - \mathbb{E}[\xi]^2.$$

It presupposes a finite mean, which finite second-order moments guarantee (previous slide); expanding the square then gives the second form. Conversely, $\mathbb{E}[\xi^2] = \text{Var}[\xi] + \mathbb{E}[\xi]^2$: a random vector has finite second-order moments **if and only if** every component has a finite variance.

For two such variables, $\text{Cov}[\xi, \eta] = \mathbb{E} [(\xi - \mathbb{E}[\xi])(\eta - \mathbb{E}[\eta])]$ is finite too, by Cauchy–Schwarz below; the covariance matrix $\Sigma = \mathbb{E} [(\xi - \mathbb{E}[\xi])(\xi - \mathbb{E}[\xi])^T]$ of a random vector is symmetric positive semi-definite.

Why the course needs exactly this. By **Cauchy–Schwarz**,

$$\mathbb{E} [|\mathbf{u}^T \mathbf{v}|] \leq \mathbb{E} [\|\mathbf{u}\| \|\mathbf{v}\|] \leq (\mathbb{E} \|\mathbf{u}\|^2)^{1/2} (\mathbb{E} \|\mathbf{v}\|^2)^{1/2}.$$

In deck 03 the recourse cost is $q(\xi)^T z(\xi)$, $z = h - Tx$, and ξ gathers the random entries of q , h , T . Finite second-order moments therefore make $\mathbb{E}$ of that product finite — which is precisely what makes $\mathcal{Q}(x)$ finite, and what the simple-recourse proof there invokes.

Independence

ξ and η are **independent** if

$$P[\xi \in A, \eta \in B] = P[\xi \in A] P[\eta \in B]$$

for all measurable A, B ; equivalently, their joint cdf factorizes.

Then $\mathbb{E}[\xi\eta] = \mathbb{E}[\xi]\mathbb{E}[\eta]$ and $\text{Var}[\xi + \eta] = \text{Var}[\xi] + \text{Var}[\eta]$.

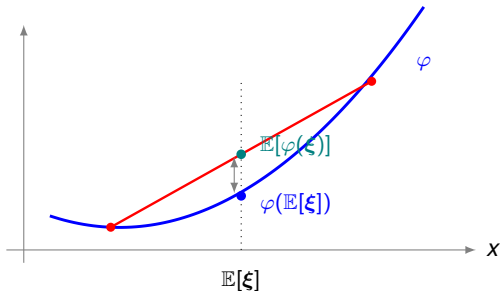
The converse fails: $\text{Cov}[\xi, \eta] = 0$ does not imply independence. Uncorrelated is weaker than independent, except in the Gaussian case where the two coincide.

Independence is what justifies $\text{Var}[\bar{X}_n] = \sigma^2/n$ for a Monte Carlo average, hence the $1/\sqrt{n}$ rate below. It is also what **fails** for the correlated estimators of deck 10, where common random numbers are used deliberately to create a positive correlation and cancel variance in a difference.

Jensen's inequality

Theorem

If φ is convex and ξ is integrable, then $\varphi(\mathbb{E}[\xi]) \leq \mathbb{E}[\varphi(\xi)]$. For φ concave the inequality is reversed.



With ξ taking two values, $\mathbb{E}[\varphi(\xi)]$ sits on the chord and $\varphi(\mathbb{E}[\xi])$ on the curve, below it. Deck 03 obtains $EV \leq WS$ this way: the value function of a linear program is convex in its right-hand side, so replacing the latter by its mean can only look *too good* — as does using $\mathbb{E}[\xi]$, if h is affine.

“Almost surely”, and why it is not a formality

An event holds **almost surely** (a.s.), or for **almost every** ξ (a.e.), if it holds on a set of probability one — it may fail, but only on a set of probability zero.

This is not the same as *always*, and deck 03 turns the difference into a worked example:

$$K_2 = \{x \mid Q(x, \xi) < \infty \text{ a.s.}\} \quad \text{versus} \quad K_2^P = \bigcap_{\xi \in \Xi} K_2(\xi) \quad (\text{every } \xi).$$

A first-stage decision may be infeasible for some realizations, provided they together have probability zero: then $x \in K_2$ but $x \notin K_2^P$.

With a **fixed** recourse the two coincide, because $\{\xi \mid x \in K_2(\xi)\}$ is then closed, and a closed set of probability one contains the support. Drop the fixed recourse and they separate — see the triangular-distribution exercise of deck 03.

Conditional expectation

For a discrete η with $P[\eta = e] > 0$,

$$\mathbb{E}[\xi \mid \eta = e] = \sum_k \xi_k P[\xi = \xi_k \mid \eta = e], \quad P[A \mid B] = \frac{P[A \cap B]}{P[B]}.$$

In general $\mathbb{E}[\xi \mid \mathcal{G}]$ is defined for a sub- σ -algebra $\mathcal{G} \subseteq \mathcal{F}$: it is the best prediction of ξ given the information in $\mathcal{G}$.

Two properties carry the multistage chapters:

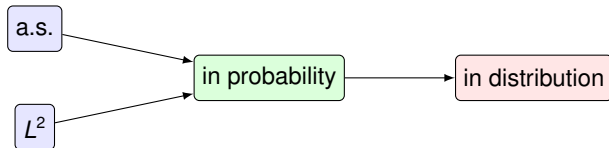
- **tower property**: $\mathbb{E}[\mathbb{E}[\xi \mid \mathcal{G}]] = \mathbb{E}[\xi]$ — this is what lets a multistage problem be written recursively, stage by stage;
- $\mathbb{E}[\xi \mid \mathcal{G}] = \xi$ when ξ is $\mathcal{G}$ -measurable, i.e. when $\mathcal{G}$ already determines ξ .

In decks 06–08, $\mathcal{G}$ is the information available at a given stage. A decision that is $\mathcal{G}$ -measurable is one that uses only what has been observed so far: that is precisely the **non-anticipativity** constraint.

Modes of convergence

For random variables X_n and X :

- **almost sure**: $P[\lim_n X_n = X] = 1$; written $X_n \rightarrow X$ a.s.
- **in probability**: $P[|X_n - X| > \epsilon] \rightarrow 0$ for every $\epsilon > 0$.
- **in L^2** (mean square): $\mathbb{E}[(X_n - X)^2] \rightarrow 0$.
- **in distribution**: $F_{X_n}(x) \rightarrow F_X(x)$ at every continuity point of F_X ; written $X_n \xrightarrow{\mathcal{D}} X$ or $X_n \Rightarrow X$.



no arrow points backwards
(except towards a constant limit)

Convergence in distribution to a *constant* does imply convergence in probability — the one reverse implication the course relies on.

The law of large numbers

Theorem (Strong law)

If $X_1, X_2, \dots$ are i.i.d. with $\mathbb{E}[|X_1|] < \infty$, then

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \longrightarrow \mathbb{E}[X_1] \quad \text{almost surely.}$$

With $X_i = g(x, \xi^i)$ and $\xi^1, \xi^2, \dots$ an i.i.d. sample, this is the whole justification of **Monte Carlo**: $\hat{g}_n(x) \rightarrow \mathbb{E}[g(x, \xi)]$ a.s.

But this is pointwise in x , and optimization needs more: the sample average must converge *uniformly* over the decision set, otherwise the minimizers need not converge. That stronger statement is the **uniform law of large numbers**,

$$\sup_{z \in S} |\hat{g}_n(z) - g(z)| \longrightarrow 0 \quad \text{a.s.,}$$

and it is the hypothesis on which the whole convergence theory of deck 09 rests.

The central limit theorem

Theorem

If $X_1, X_2, \dots$ are i.i.d. with mean μ and **finite** variance σ^2 , then

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{\mathcal{D}} N(0, \sigma^2), \quad \text{i.e.} \quad \bar{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right) \text{ for large } n.$$

The law of large numbers says the estimator converges; the central limit theorem says **how fast**, and with which spread.

Two consequences used throughout the course:

- the standard error is $\sigma/\sqrt{n}$: to **halve** it, the sample size must be **quadrupled**. The $\sqrt{n}$ is not an artefact of the analysis, it is the rate;
- the rate does not depend on the dimension of ξ — which is why sampling survives where scenario enumeration explodes, as the 7^{86} example of deck 03 shows.

Estimators and confidence intervals

The sample average $\bar{X}_n$ is an **unbiased** estimator of μ : $\mathbb{E}[\bar{X}_n] = \mu$, with $\text{Var}[\bar{X}_n] = \sigma^2/n$ when the X_i are independent. The variance σ^2 is itself estimated by

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

(the $n - 1$ makes it unbiased).

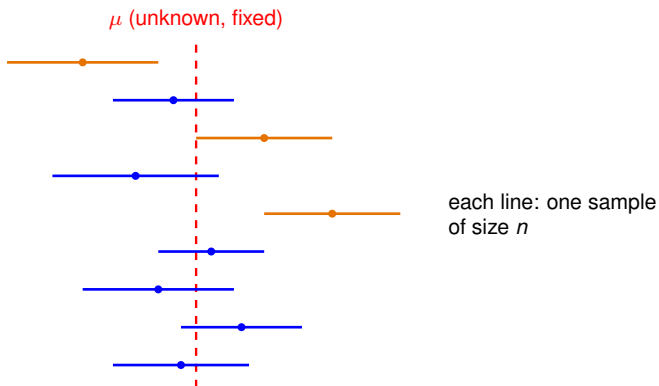
The central limit theorem then gives an **asymptotic confidence interval** at level $1 - \alpha$:

$$\bar{X}_n \pm \Phi^{-1}\left(1 - \frac{\alpha}{2}\right) \frac{S_n}{\sqrt{n}},$$

where Φ is the cdf of $N(0, 1)$. For $\alpha = 0.05$, $\Phi^{-1}(0.975) \approx 1.96$; for $\alpha = 0.1$, $\Phi^{-1}(0.95) \approx 1.64$.

This is the exact formula used in the stopping rule of deck 07 and the error estimates of deck 10. Two warnings: the interval is **asymptotic**, so it is unreliable for small n ; and it requires **independent** draws, which the batch means of deck 12 are designed to restore when the observations are correlated.

Why the interval is a statement about the method



“A 95% confidence interval” does **not** mean μ has a 95% chance of lying in the interval you computed: μ is not random, and your interval is already drawn. It means that the *procedure*, repeated on fresh samples, produces an interval covering μ in 95% of the cases. The orange lines are the unlucky 5%.

The distributions the course uses

Distribution	Density / mass	Mean, variance	Used in
$U[a, b]$	$1/(b-a)$ on $[a, b]$	$\frac{a+b}{2}, \frac{(b-a)^2}{12}$	deck 02 (generators)
$N(\mu, \sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$	μ, σ^2	decks 05, 07, 10
$\text{Exp}(\lambda)$	$\lambda e^{-\lambda x}, x \geq 0$	$1/\lambda, 1/\lambda^2$	deck 12 (inter-arrivals)
$\text{Poisson}(\lambda)$	$e^{-\lambda} \lambda^k / k!$	λ, λ	deck 12 (arrival counts)
χ_k^2	—	$k, 2k$	deck 12 (variance CIs)

Two facts worth remembering:

- the exponential is **memoryless**, $P[\xi > s + t \mid \xi > s] = P[\xi > t]$, and it is the only continuous distribution that is. This is why exponential inter-arrivals make the queueing models of deck 12 tractable;
- exponential inter-arrivals and Poisson counts describe the **same** process, seen through time or through counts.

The normal distribution, and the multivariate case

Φ denotes the cdf of $N(0, 1)$; it has no closed form, and Φ^{-1} is evaluated numerically. If $\xi \sim N(\mu, \sigma^2)$ then $(\xi - \mu)/\sigma \sim N(0, 1)$, so

$$P[\xi \leq t] = \Phi\left(\frac{t - \mu}{\sigma}\right).$$

In $\mathbb{R}^n$, $\xi \sim N(\mu, \Sigma)$ with Σ symmetric positive definite has density

$$f(x) = \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right),$$

and **any affine image is again normal**: $a^T \xi \sim N(a^T \mu, a^T \Sigma a)$.

This last fact is the whole mechanism of deck 05. A chance constraint $P[a^T \xi \leq b] \geq \alpha$ becomes, after standardization,

$$\Phi^{-1}(\alpha) \sqrt{a^T \Sigma a} + a^T \mu \leq b,$$

a **deterministic** — and, for $\alpha \geq 1/2$, convex — constraint. Outside the Gaussian case no such closed form exists, which is why that deck turns to log-concavity instead.

Notation across the decks

The slides are not always uniform; the variants you will meet are the following.

Object	Variants used
Random vector / realization	ξ / ξ , occasionally ξ
Probability	$P[\cdot]$, $\mathbb{P}[\cdot]$
Expectation	$\mathbb{E}[\cdot]$, $\mathbb{E}_{\xi}[\cdot]$
Convergence in distribution	$\xrightarrow{\mathcal{D}}$ (deck 07), $\Rightarrow$ (deck 10)
Normal law	$N(\mu, \sigma^2)$, $\mathcal{N}(\mu, \sigma^2)$
Normal quantile	$\Phi^{-1}(1 - \alpha/2)$ (deck 07), α_{δ} (deck 10)

They all denote the same objects. The subscript in $\mathbb{E}_{\xi}$ is only there to recall **which** variable is being averaged out — useful as soon as two sources of randomness coexist, as in the sampling chapters.

References

- P. Billingsley, *Probability and Measure*, 3rd edition, Wiley, 1995 — the measure-theoretic reference.
- G. Grimmett and D. Stirzaker, *Probability and Random Processes*, 3rd edition, Oxford University Press, 2001 — a gentler route to the same material.
- A.W. van der Vaart, *Asymptotic Statistics*, Cambridge University Press, 1998 (Chapters 2–3, for the modes of convergence and the central limit theorem).
- S.M. Ross, *Simulation*, 5th edition, Academic Press, 2013 — estimators, confidence intervals and variance reduction, as used in decks 10 and 12.
- J.R. Birge and F. Louveaux, *Introduction to Stochastic Programming*, 2nd edition, Springer, 2011 (Chapter 2, for the probabilistic setting of two-stage models).